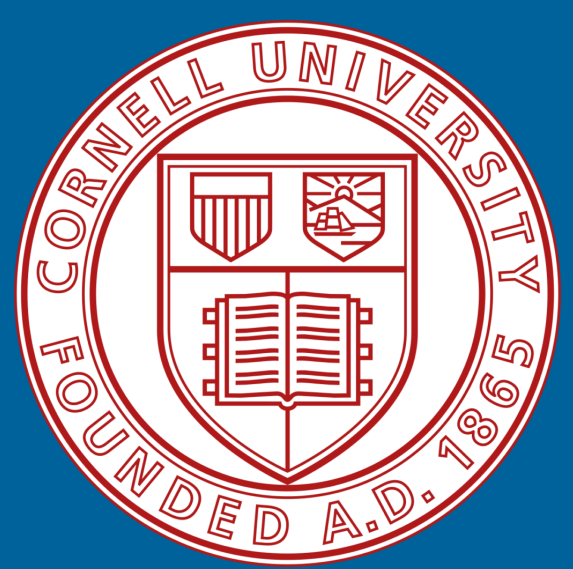
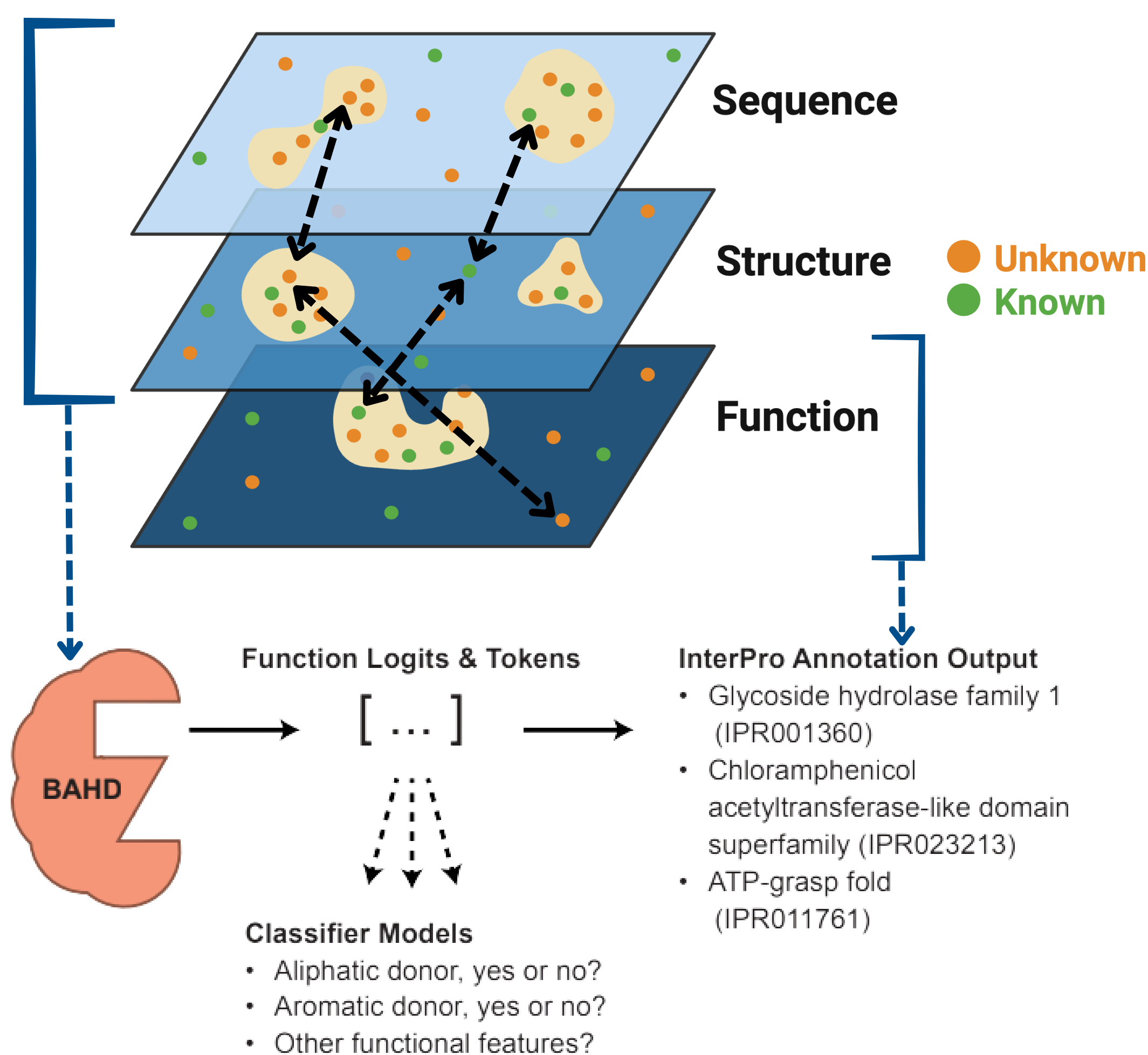




Transfer learning of ESM-3 for enzyme substrate prediction

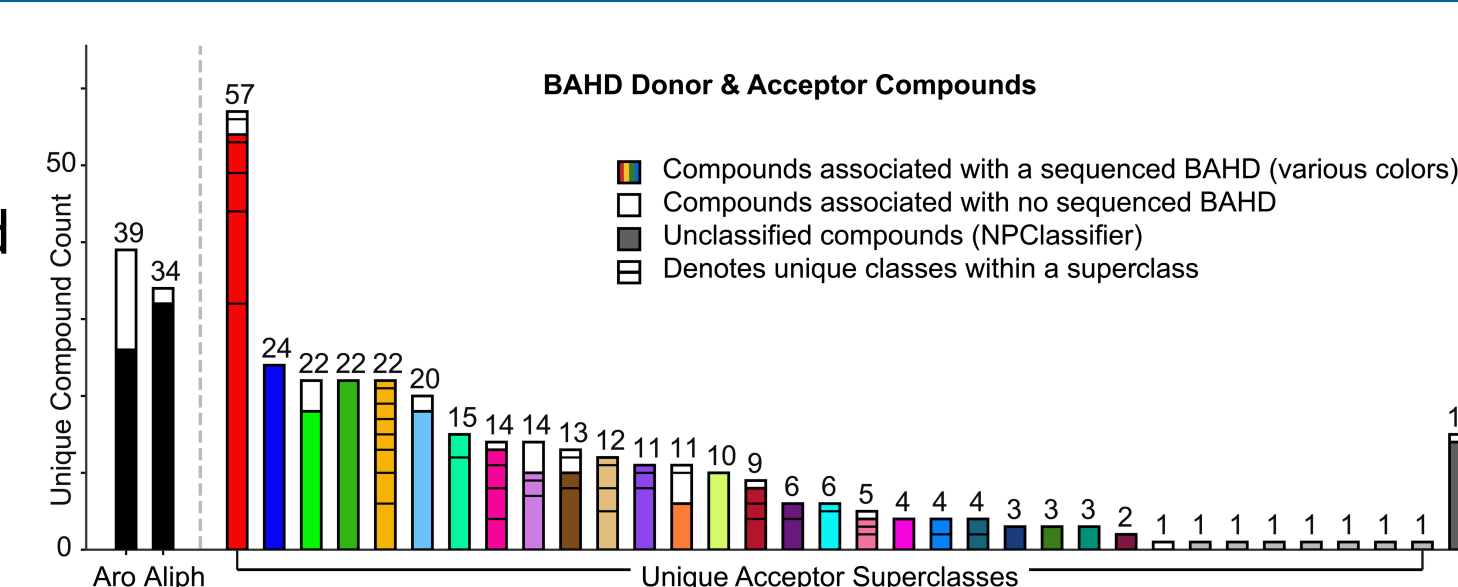
Nate Smith (nss97@cornell.edu), Alen Zimic (amz63@cornell.edu), Chase Holdener (ch2228@cornell.edu), Darke Hull (drh257@cornell.edu)

ESM3's function-prediction logits are too coarse to resolve donor/acceptor specificity in BAHD enzymes, collapsing them into the same broad functional label.

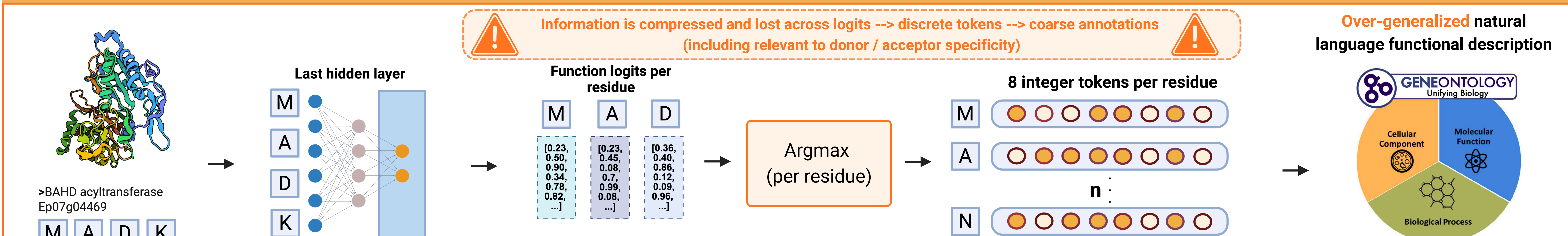


Dataset

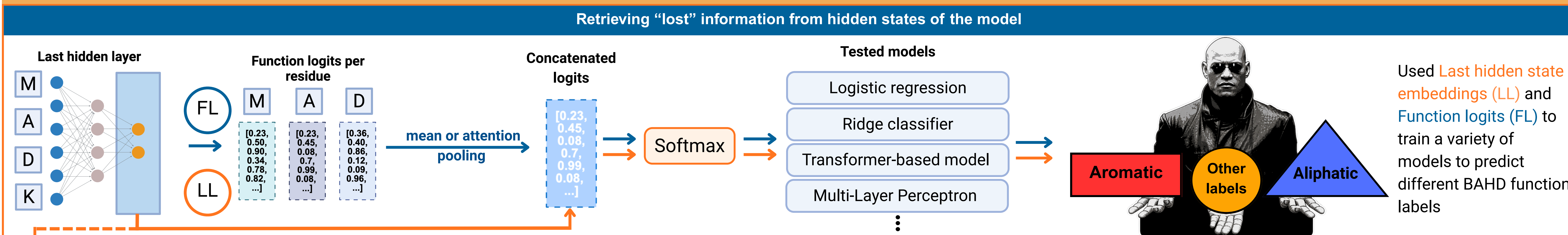
The FuncFetch workflow (Smith et al. 2025) retrieved thousands of enzyme activities. We focused on the **BAHD** (366 sequences) and **UGT** families (482 sequences), each with literature backed substrate labels.



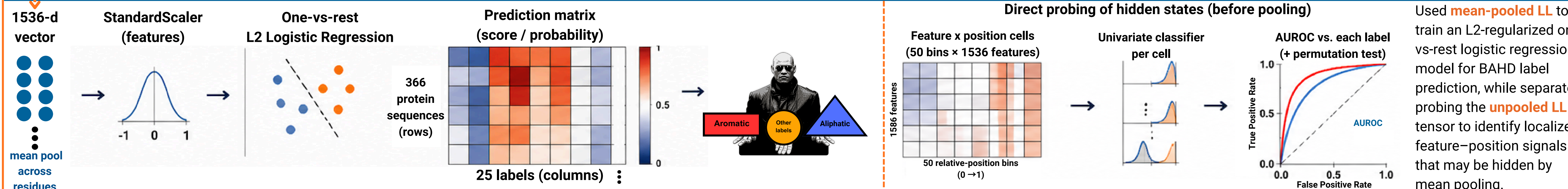
ESM-3 ORIGINAL WORKFLOW



ADAPTED WORKFLOW

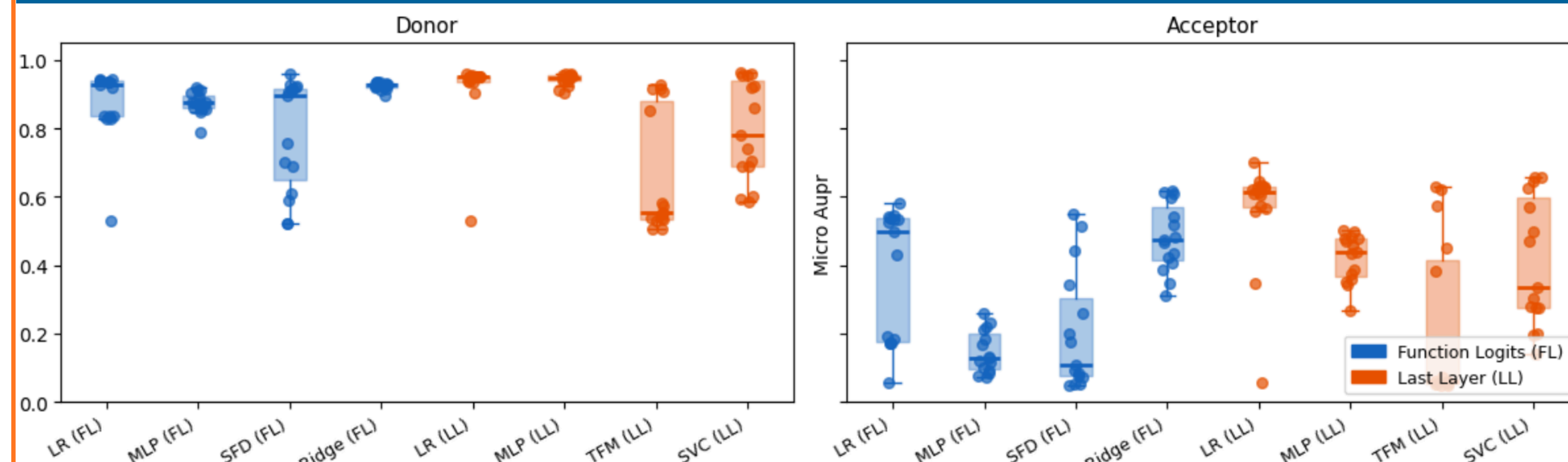


Individual ESM3 features per residues are important for acceptor/donor classification task

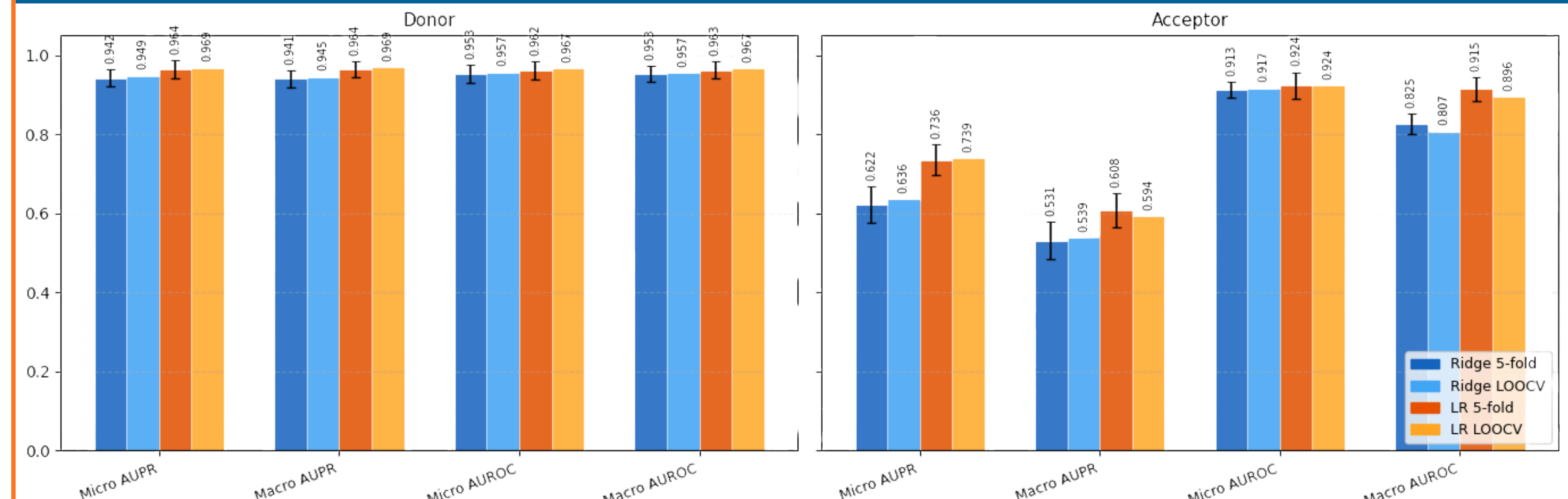


RESULTS

5-fold average of 15 random search model screening runs



Performance of selected FL and LL models after hyper-parameter tuning

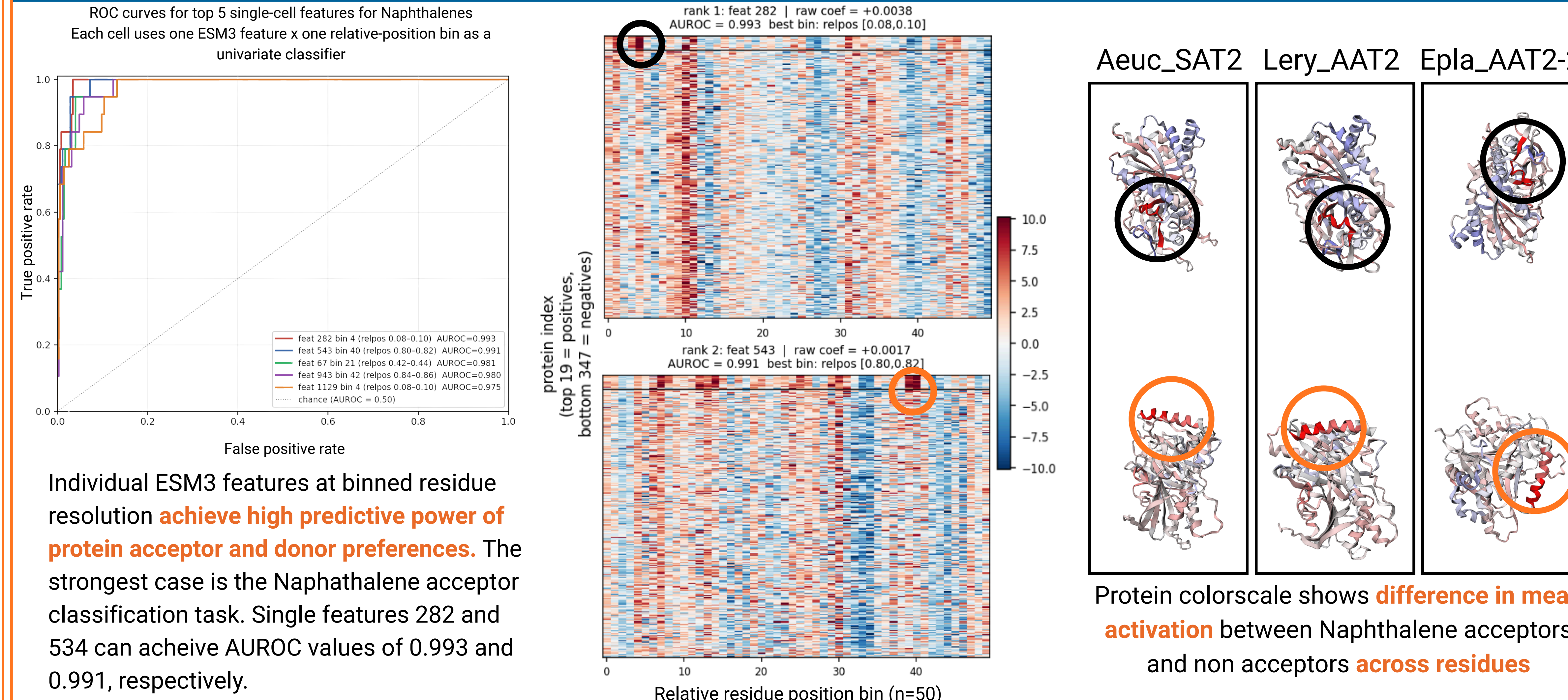


Optimized on acceptor micro-AUPR because **acceptor specificity was the hardest** and most fine-grained task.

Across 15 random-search runs, top FL/LL feature-model combinations showed **similar performance trends**.

Trends were consistent across 5-fold CV and LOOCV, supporting **split-robust generalization**.

Localized ESM3 Features Predict Naphthalene Acceptor Specificity



CONCLUSIONS

Mean-pooled ESM-3 last-layer embeddings support accurate BAHD donor/acceptor prediction, but their interpretations are diffuse when residues or features are analyzed separately.

Direct probing of feature x residue-position bins reveals strong localized substrate-class signal, in some cases, a single ESM-3 feature at one relative protein region separates acceptors from non-acceptors with near-perfect

ESM-3 embeddings encode BAHD donor and acceptor specificity in both global representations and localized last-layer feature-position signals, enabling lightweight prediction and interpretability without full model fine-tuning.

ACKNOWLEDGEMENTS

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REFERENCES

Adapted from Hayes et al., Science, 2025